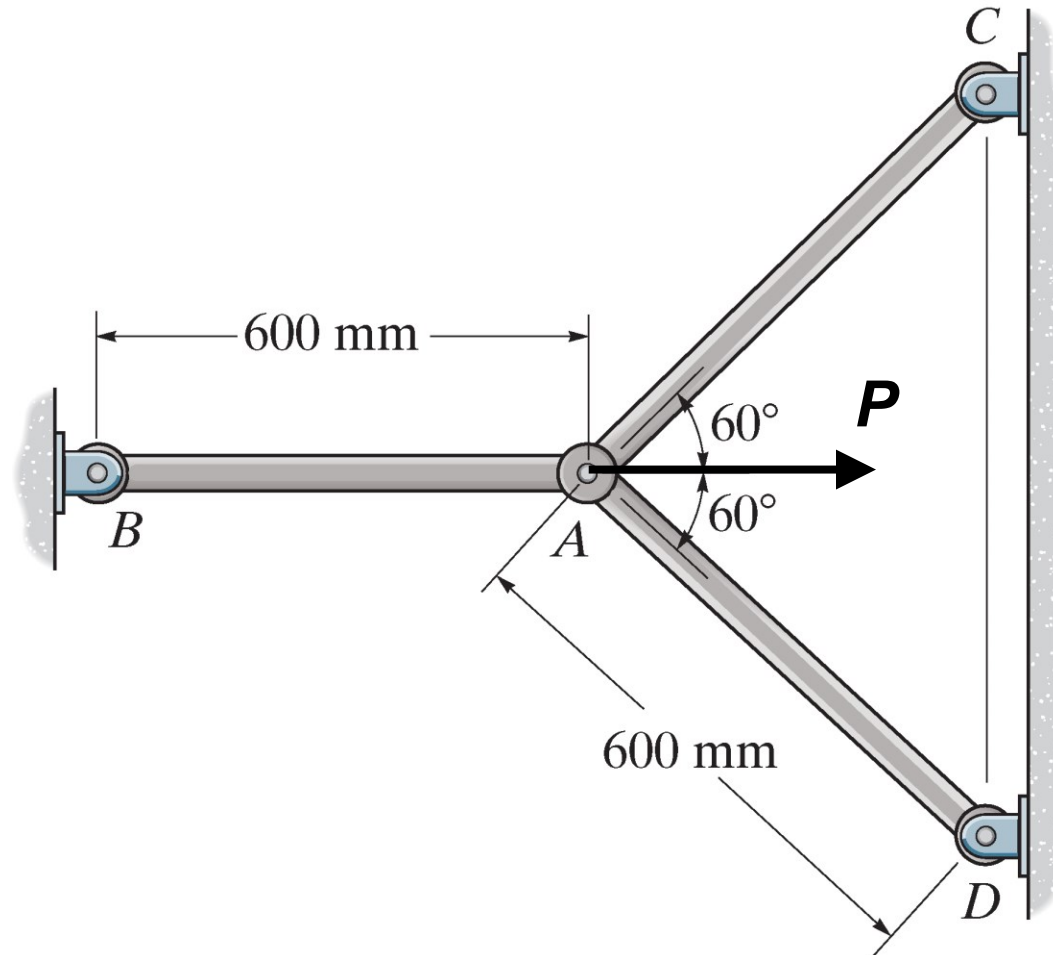


1. The rods each have the same 20-mm diameter and 600-mm length. The modulus of elasticity of all rods is the same ($E = 200 \text{ GPa}$). Determine the force developed in each rod when the external force $P = 300 \text{ kN}$ is applied horizontally at joint A . (10')



Solutions to Extra Exercise #1:

Assuming the xy -axis and also the resultant force in the bar as shown in the figure, for the joint A , the free-body diagram gives:

$$\sum F_x = 0 : P + F_{AC} \cos 60^\circ + F_{AD} \cos 60^\circ - F_{AB} = 0 \quad (1)$$

$$\sum F_y = 0 : F_{AC} \sin 60^\circ - F_{AD} \sin 60^\circ = 0 \quad (2a)$$

$$\rightarrow F_{AC} = F_{AD} \quad (2b)$$

Substituting (2b) into (1), we have:

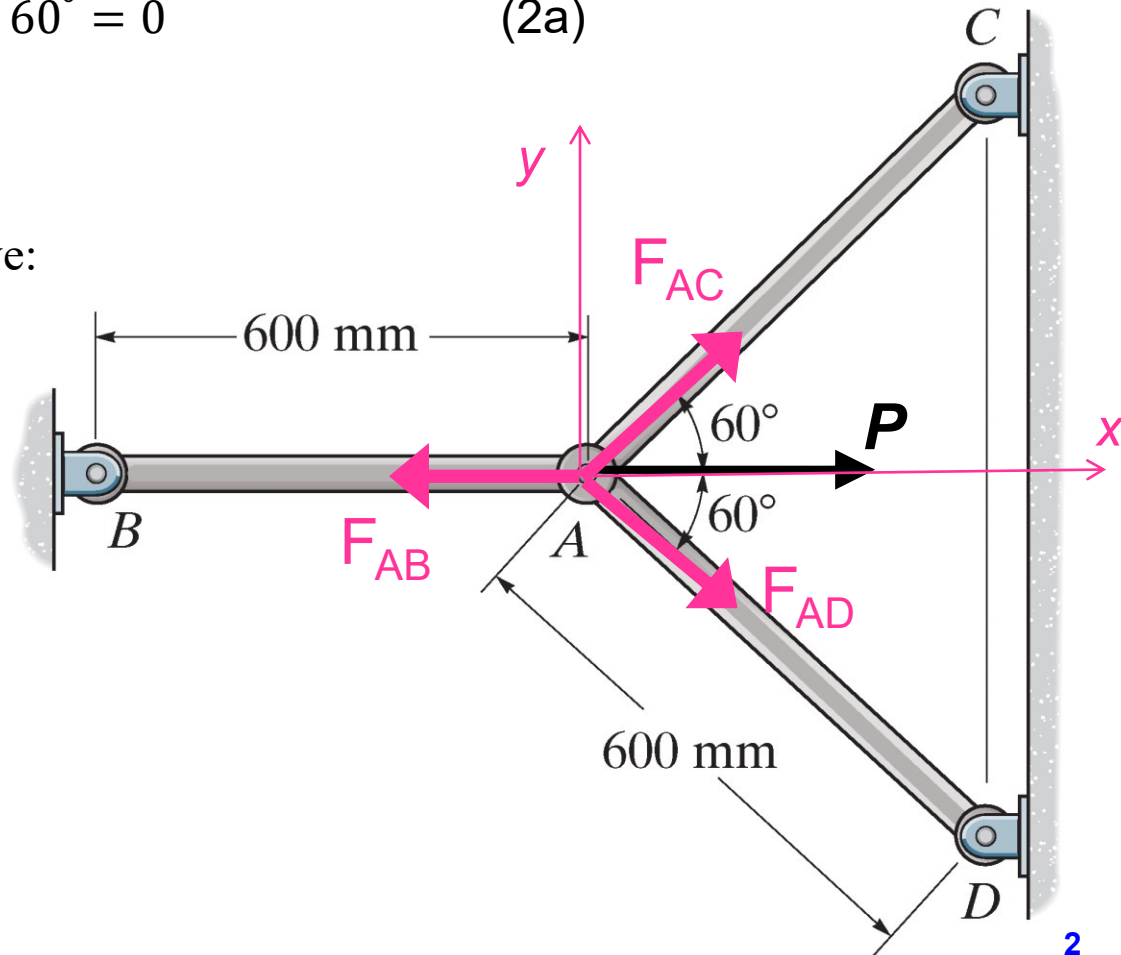
$$P + F_{AC} - F_{AB} = 0 \quad (3)$$

Compatibility condition:

The absolute value of horizontal displacement of bar AB should be equal to the horizontal part of compression on bar AC (or AD).

$$\delta_{AB} + \delta_{AC} \sin 30^\circ = 0$$

$$\rightarrow \delta_{AB} = -0.5\delta_{AC} \quad (4)$$



Solutions to Extra Exercise #1, cont'd:

The elongation / compression of the bar can be obtained by the force (related to stress) and modulus of elasticity (refer to lecture ppt Ch.4.1 page 7):

$$\delta_{AB} = \frac{F_{AB}L_{AB}}{EA}$$

$$\delta_{AC} = \frac{F_{AC}L_{AC}}{EA}$$

Since the modulus, cross-sectional area, and the length of AB and AC are the same, substituting these into Eq. (4), we get

$$F_{AB} = -0.5F_{AC} \quad (5)$$

Substituting (5) into (3), we have

$$P + 1.5F_{AC} = 0 \quad \rightarrow \quad F_{AC} = -\frac{P}{1.5} = -200 \text{ kN} \quad (\text{Ans.})$$

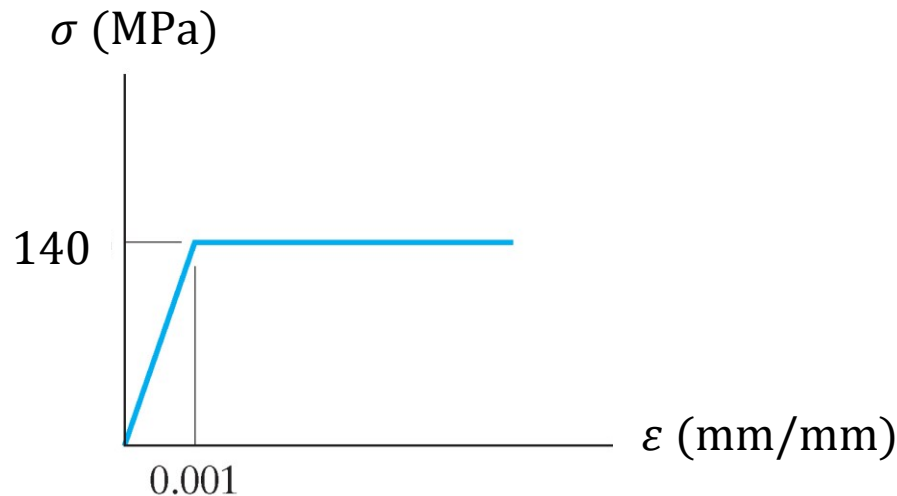
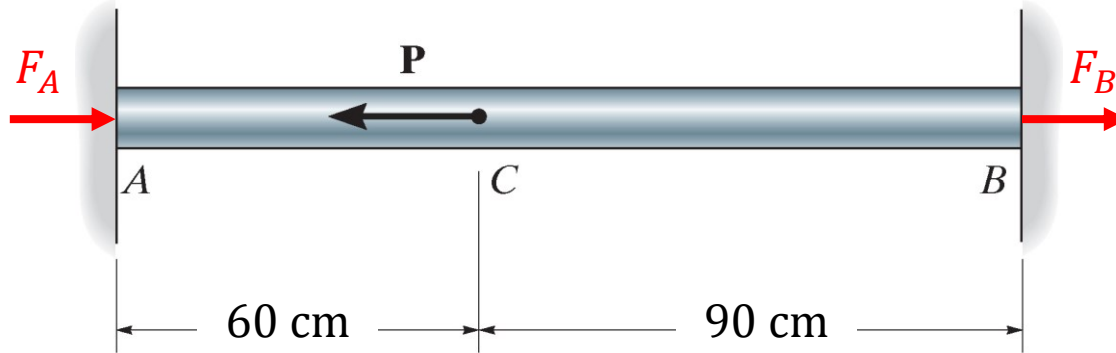
$$\text{According to (3b) and (5), we have} \quad F_{AB} = -0.5F_{AC} = 100 \text{ kN} \quad (\text{Ans.})$$

$$F_{AD} = F_{AC} = -200 \text{ kN} \quad (\text{Ans.})$$

The negative sign indicates the assumed direction of the forces should be opposite.

2. The bar having a diameter of 5 cm is fixed connected at its ends and supports the axial load P . If the material is idealized elastoplastic as shown by the stress-strain diagram, determine the smallest load P needed to cause segment CB to yield. If this smallest load is released, determine the displacement of point C .

(25')



Solutions to Extra Exercise #2:

Step 1: Equilibrium

Assuming the resultant reactions at the two ends A and B as shown in the figure, for the entire bar, the free-body diagram (force equation of equilibrium) gives:

$$\sum F_x = 0 \rightarrow$$

$$F_A + F_B = P \quad (1)$$

Step 2: Compatibility Condition

The total strain or displacement of the entire bar (from A to B) should be zero, i.e.

$$\delta_{A/B} = 0 \rightarrow \delta_{AC} + \delta_{CB} = 0 \quad (2)$$

Solutions to Extra Exercise #2, cont'd:

Step 3: Load-Displacement Relations

The elongation / compression of the bar can be obtained by the force (related to stress) and modulus of elasticity (refer to lecture ppt Ch.4.1 page 7):

$$\delta_{AC} = - \frac{F_A L_{AC}}{EA}$$

$$\delta_{CB} = + \frac{F_B L_{CB}}{EA}$$

Substituting them into Eq. (2), we get

$$- \frac{F_A L_{AC}}{EA} + \frac{F_B L_{CB}}{EA} = 0$$

$$\rightarrow F_A L_{AC} = F_B L_{CB}$$

$$\rightarrow F_A = 1.5 F_B \quad (3)$$

Combining Eqs. (1) and (3), we obtain:

$$F_A = 0.6P \quad F_B = 0.4P$$

Since $F_A > F_B$, it means that, when P is increased, **region AC will become plastic first**, then CB will become plastic. Therefore, the minimum load P needed to cause segment CB to yield is

$$F_B = 0.4P = \sigma_Y \cdot A = 140 \times 10^6 \times \frac{\pi}{4} \times 0.05^2 = 274.89 \text{ kN}$$

$$\rightarrow P = 687.2 \text{ kN} \quad (\text{Ans.})$$

Solutions to Extra Exercise #2, cont'd:

Step 4: Unloading

From above analysis, we know that, when loading $P = 687.2$ kN is applied, **region AC will become plastic first** and then CB will become plastic. The strain in segment CB can be determined by the Yield strength as shown in the stress-strain diagram of the elastoplastic material. Note that the both ends are fixed, then the total (compressive) elongation in segment AC (already in plastic region) should be equal to the (tensile) elongation in CB .

$$\varepsilon_{CB} = 0.001 \text{ mm/mm}$$

$$\delta_{AC} = \delta_{CB} = \varepsilon_{CB} \cdot L_{CB} = 0.001 \times 90 = 0.09 \text{ cm}$$

$$\varepsilon_{AC} = \frac{\delta_{AC}}{L_{AC}} = \frac{0.09}{60} = 0.0015 \text{ mm/mm} > 0.001 \text{ mm/mm (AC in plastic region)}$$

When the loading is released, the segment CB will try to recover to its original (undeformed) state along the Young's modulus slop line. However, since CB is connected with AC and AC already reaches the plastic region, the AC will not go back to the original undeformed state. Instead, the AC will first recover all compressive stress back to zero (leaving permanent elongation, point C_p in the figure) and then will be pulled further due to the residual tensile stress in CB that cannot be fully recovered back to zero (because of the permanent set in AC).

Solutions to Extra Exercise #2, cont'd:

Step 4: Unloading, cont'd

First, we get plastic strain in AC and its permanent set

$$\varepsilon_{pl} = \varepsilon_{AC} - \varepsilon_Y = 0.0015 - 0.001 = 0.0005 \text{ mm/mm}$$

$$\delta_m = \varepsilon_{pl} L_{AC} = 0.0005 \times 60 = 0.03 \text{ cm}$$

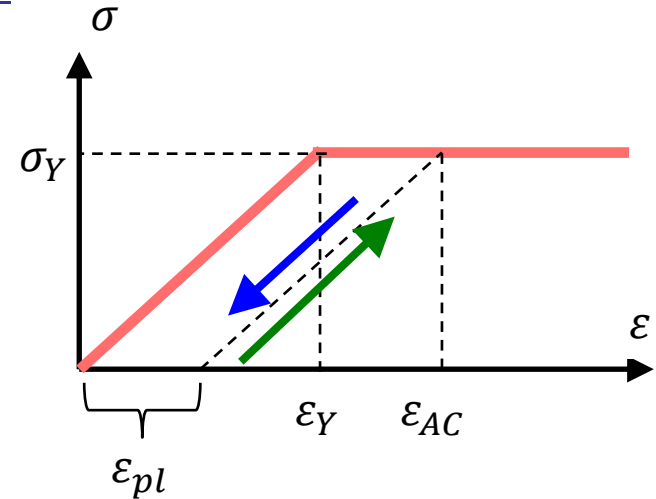
C_0 is the original position of point C before loading

C' is the position of point C after loading is applied

C_p is the position of point C after the stress in AC region goes back to zero

C_{final} is the final position of point C after **the stress in AC is in balance with that in AB**
(both regions are tensile stress)

AC is first unloaded (blue arrow) and then loaded again with tensile stress (green arrow)!



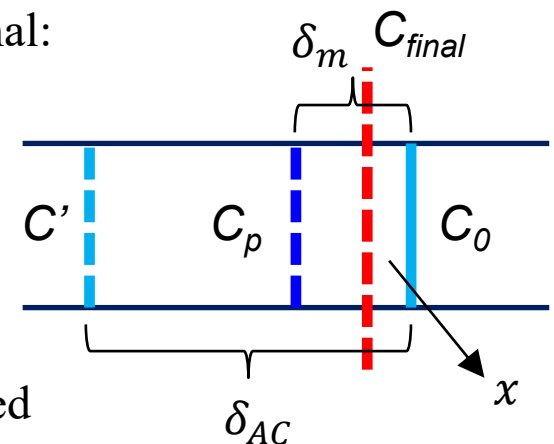
Assume the final position of point C is in x relative to the original:

$$N_{CB} = N_{AC}$$

$$\rightarrow E \cdot A \cdot \frac{x}{90} = E \cdot A \cdot \frac{\delta_m - x}{60}$$

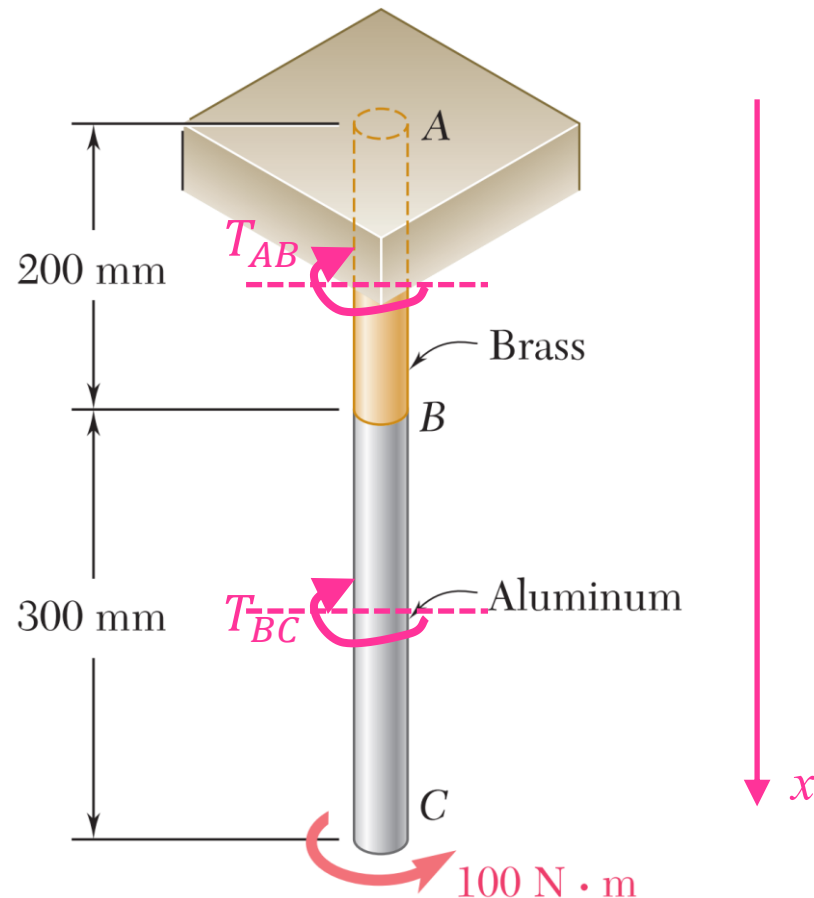
$$\rightarrow x/90 = (0.03 - x)/60$$

$$\rightarrow x = 0.018 \text{ cm} \quad (\text{Ans.})$$



The displacement of point C with respect to the original unloaded position is 0.018 cm to the left.

3. The aluminum rod BC (shear modulus $G = 26 \text{ GPa}$) is bonded to the brass rod AB ($G = 39 \text{ GPa}$) and then is fixed to the wall at the end A . Knowing that each rod is solid and has a diameter of 12 mm , determine the angle of twist (a) at B with respect to A , (b) at C with respect to A . (10')



Solutions to Extra Exercise #3:

Step 1: Equilibrium

For segment AB and BC , assuming the resultant internal moment as shown in the figure, apply the free-body diagram:

$$\sum M = 0 \rightarrow$$

$$T_{BC} = -100 \text{ N} \cdot \text{m}$$

$$T_{AB} = -100 \text{ N} \cdot \text{m}$$

Step 2: Load-Displacement Relations

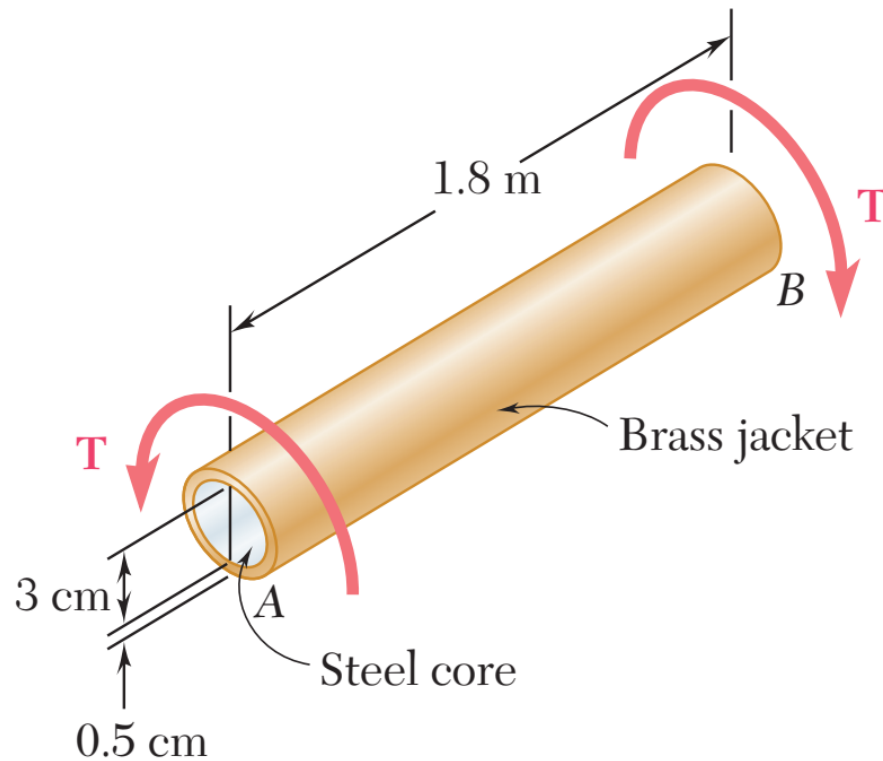
The angle of twist for each segment of the shaft can be obtained by the torque, shear modulus of elasticity, and the polar moment of inertia:

$$\phi_{A/B} = \frac{T_{AB}L_{AB}}{JG} = -\frac{100 \times 0.2}{\frac{\pi}{2} \times 0.006^4 \times 39 \times 10^9} = -0.2519 \text{ rad} \quad (\text{Ans.})$$

$$\phi_{A/C} = \frac{T_{AB}L_{AB}}{JG_{br}} + \frac{T_{BC}L_{BC}}{JG_{al}}$$

$$= -\frac{100 \times 0.2}{\frac{\pi}{2} \times 0.006^4 \times 39 \times 10^9} - \frac{100 \times 0.3}{\frac{\pi}{2} \times 0.006^4 \times 26 \times 10^9} = -0.8187 \text{ rad} \quad (\text{Ans.})$$

4. The composite shaft shown consists of a 0.5-cm-thick brass jacket ($G_{\text{br}} = 38 \text{ GPa}$) bonded to a 3.0-cm-diameter steel core ($G_{\text{st}} = 76 \text{ GPa}$). Knowing that the shaft is subjected to $600 \text{ N} \cdot \text{m}$ torques, determine (a) the maximum shearing stress in the brass jacket, (b) the maximum shearing stress in the steel core, (c) the angle of twist of end B relative to end A . (20')



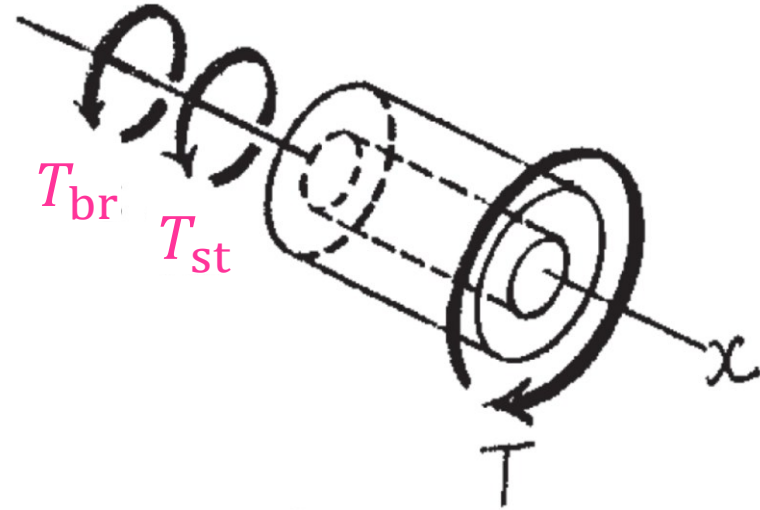
Solutions to Extra Exercise #4:

Step 1: Equilibrium

Assuming the resultant internal torques in the brass jacket and steel core as T_{br} and T_{st} , respectively. For the entire specimen, the free-body diagram (moment equation of equilibrium) gives:

$$\sum M = 0 \rightarrow$$

$$T_{br} + T_{st} = T = 600 \quad (1)$$



Step 2: Compatibility Condition

Since the brass jacket is bonded firmly to the steel core, the angle of twist of the brass jacket and the steel core must be the same, i.e.

$$\phi_{br} = \phi_{st} \quad (2)$$

Solutions to Extra Exercise #4, cont'd:

Step 3: Load-Displacement Relations

The angle of twist for the brass jacket and the steel core can be obtained by the internal torque, shear modulus of elasticity, and the polar moment of inertia:

$$\phi_{\text{br}} = \frac{T_{\text{br}} L_{AB}}{J_{\text{br}} G_{\text{br}}}$$

$$\phi_{\text{st}} = \frac{T_{\text{st}} L_{AB}}{J_{\text{st}} G_{\text{st}}}$$

Substituting them into Eq. (2) and considering $J_{\text{br}} = \frac{\pi}{2} (0.02^4 - 0.015^4)$, $J_{\text{st}} = \frac{\pi}{2} \times 0.015^4$, $G_{\text{br}} = 38 \text{ GPa}$, $G_{\text{st}} = 76 \text{ GPa}$, we get

$$T_{\text{br}} = 1.0802 T_{\text{st}} \quad (3)$$

Combining Eqs. (1) and (3), we get

$$T_{\text{br}} = 311.6 \text{ N} \cdot \text{m}$$

$$T_{\text{st}} = 288.4 \text{ N} \cdot \text{m}$$

Solutions to Extra Exercise #4, cont'd:

Step 4: Maximum Shearing Stress

$$(\tau_{\max})_{\text{br}} = \frac{T_{\text{br}} c_2}{J_{\text{br}}} = \frac{311.6 \times 0.02}{\frac{\pi}{2} (0.02^4 - 0.015^4)} = 36.3 \text{ MPa} \quad (\text{Ans.})$$

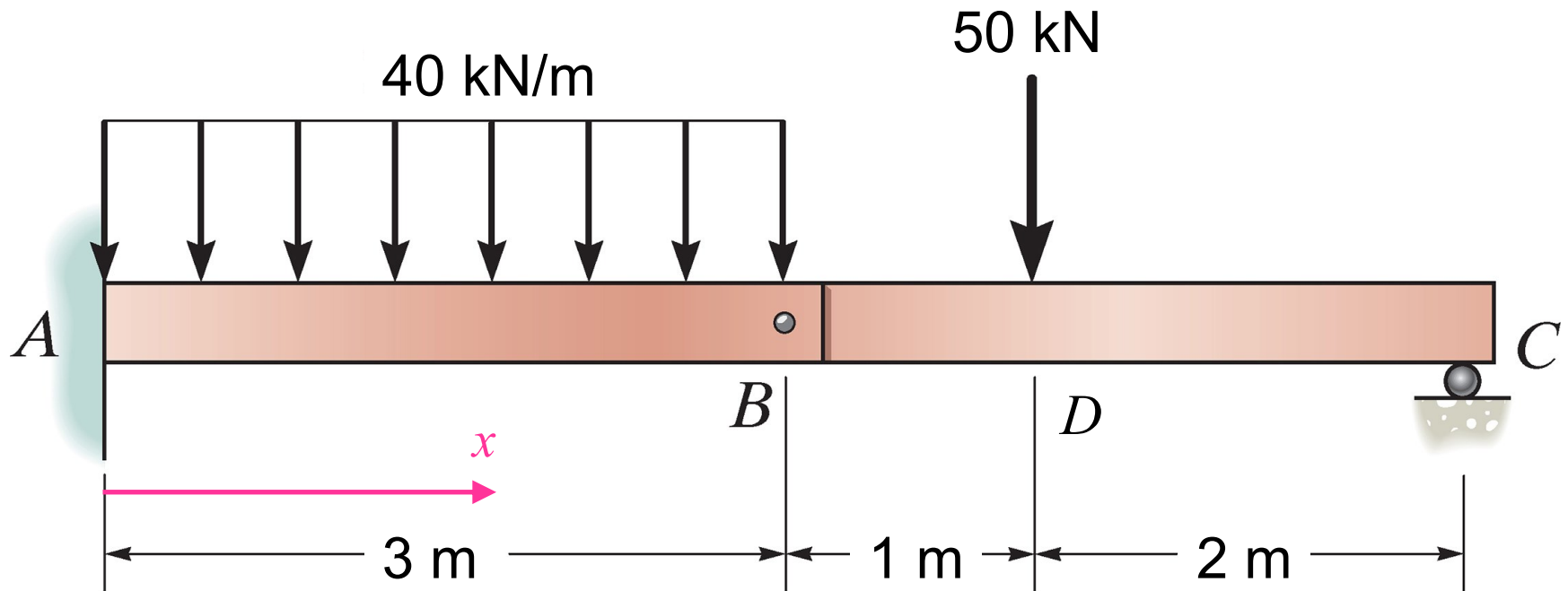
$$(\tau_{\max})_{\text{st}} = \frac{T_{\text{st}} c_1}{J_{\text{st}}} = \frac{288.4 \times 0.015}{\frac{\pi}{2} \times 0.015^4} = 54.4 \text{ MPa} \quad (\text{Ans.})$$

Step 5: Angle of Twist

The angle of twist can be obtained by either component in the specimen:

$$\phi_{AB} = \phi_{\text{st}} = \frac{T_{\text{st}} L_{AB}}{J_{\text{st}} G_{\text{st}}} = \frac{288.4 \times 1.8}{\frac{\pi}{2} \times 0.015^4 \times 76 \times 10^9} = 0.086 \text{ rad} \quad (\text{Ans.})$$

5. Determine the shear and moment for the beam as a function of position x and plot the diagrams. Note: the shear and moment diagrams can be calculated either by section method or graphical method. (15')



Solutions to Extra Exercise #5:

We use “Graphical Method” to solve this problem

Step 1: Support Reactions

Applying free-body diagram to the two beams separately:

For shaft AB:

$$\sum F_y = 0 \rightarrow$$

$$F_A - F_B = 40 \times 3 = 120 \text{ kN}$$

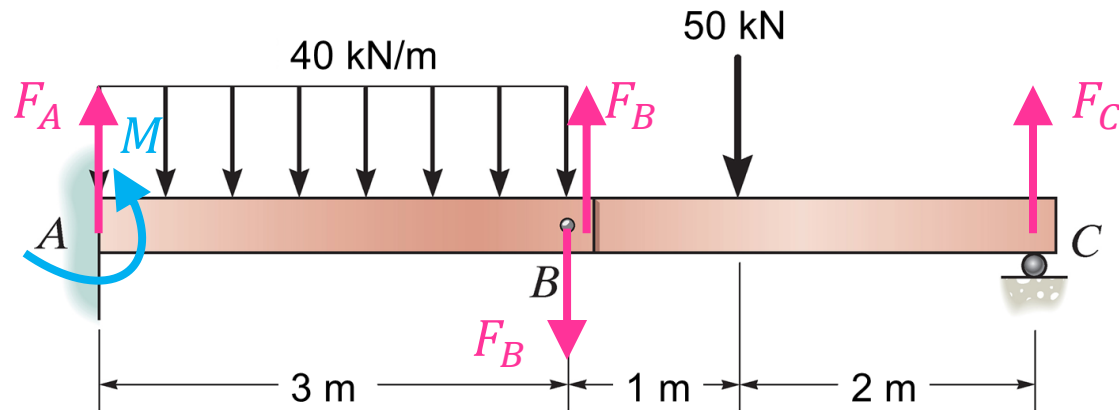
$$\sum M_A = 0 \rightarrow M = F_B \times 3 + 40 \times 3 \times 1.5$$

For shaft BC:

$$\sum F_y = 0 \quad \rightarrow \quad F_B + F_C = 50$$

$$\sum M_B = 0 \quad \rightarrow \quad F_C \times 3 = 50 \times 1$$

$$\begin{aligned} \rightarrow \quad F_A &= 153.33 \text{ kN} \\ F_B &= 33.33 \text{ kN} \\ F_C &= 16.67 \text{ kN} \\ M &= 280 \text{ kN} \cdot \text{m} \end{aligned}$$



Solutions to Extra Exercise #5, cont'd:

Step 2: Shear Diagram

Note: we need to process the two beams separately!!!

For shaft AB:

a) Determine the shear at left end first (the origin A of x -axis):

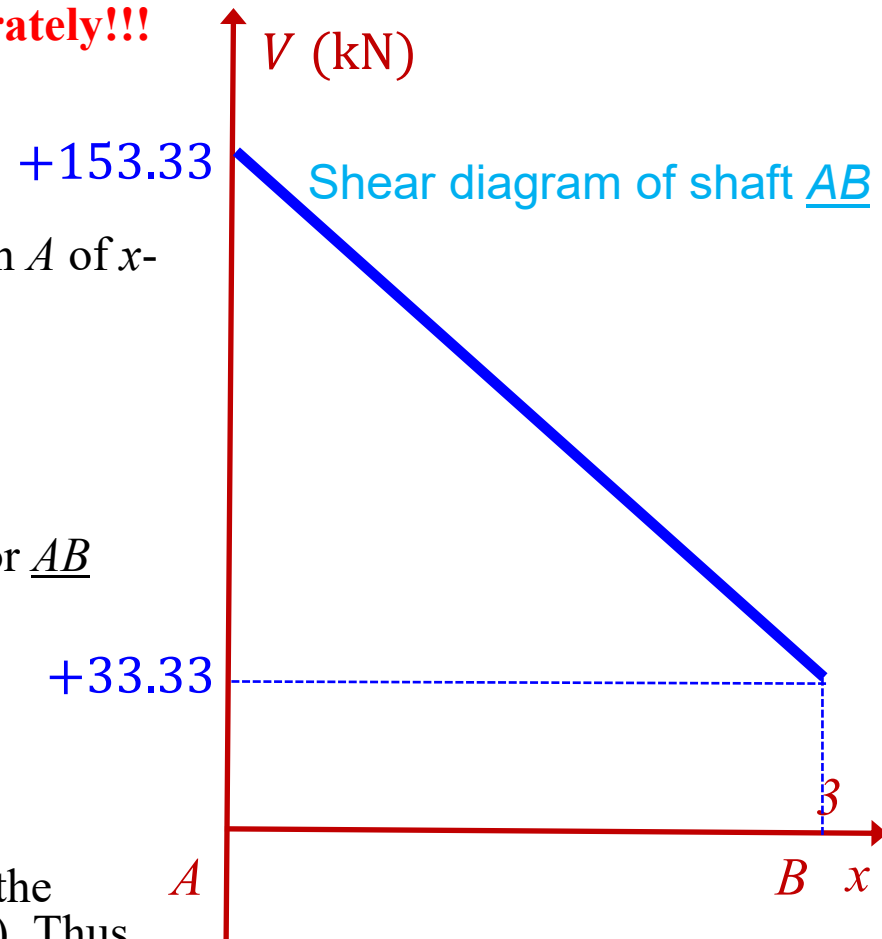
$$x = 0, \quad V = 153.33 \text{ kN}$$

b) Determine V diagram according to $\frac{dV}{dx} = w$ for $\underline{AB}$ region:

$$\frac{dV}{dx} = -40$$

$$\rightarrow V = -40x + C_1$$

The constant C_1 can be determined by applying the “**boundary**” condition ($x = 0$, $V = 153.33 \text{ kN}$). Thus, $C_1 = 153.33$ and $V = -40x + 153.33 \text{ kN}$ ($0 \leq x \leq 3 \text{ m}$).



Solutions to Extra Exercise #5, cont'd:

Step 3: Moment Diagram

For shaft AB:

a) Determine the moment at left end first (the origin A of x -axis):

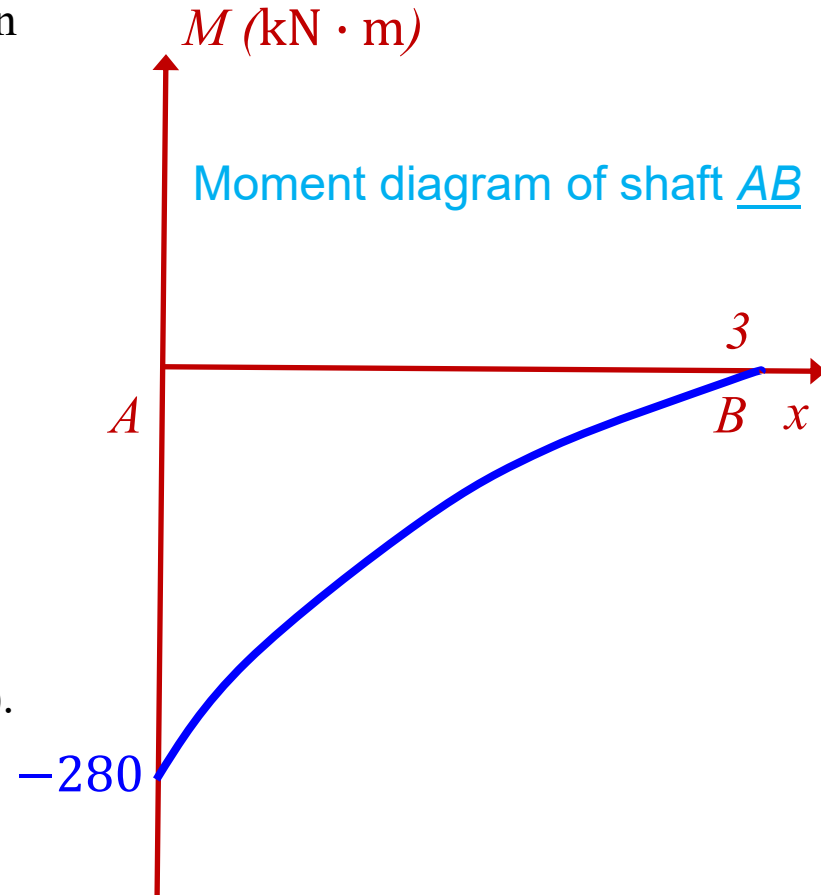
$$x = 0, \quad M = -280 \text{ kN} \cdot \text{m}$$

b) Determine **M** diagram according to $\frac{dM}{dx} = V$ for AB region:

$$\frac{dM}{dx} = -40x + 153.33$$

$$\rightarrow M = -20x^2 + 153.33x + C_4$$

The constant C_4 can be determined by applying the “**boundary**” condition ($x = 0$, $M = -280 \text{ kN} \cdot \text{m}$). Thus, $C_4 = 280$ and $M = -20x^2 + 153.33x - 280 \text{ kN} \cdot \text{m}$ ($0 \leq x \leq 3 \text{ m}$).



Solutions to Extra Exercise #5, cont'd:

Step 2: Shear Diagram, Cont'd

Note: we need to process the two beams separately!!!

For shaft BC:

a) Determine the shear at left end first (the origin B of x -axis):

$$x = 3, \quad V = +33.33 \text{ kN}$$

b) Determine V diagram according to $\frac{dV}{dx} = w$ for $\underline{BD}$ region:

$$\frac{dV}{dx} = 0$$

$$\rightarrow V = C_2$$

The constant C_2 can be determined by applying the “**boundary**” condition ($x = 3$, $V = +33.33 \text{ kN}$). Thus, $C_2 = 33.33$ and $V = 33.33 \text{ kN}$ ($3 \text{ m} \leq x \leq 4 \text{ m}$).

c) At point D , due to the external concentrated force 50 kN (downward), the V decreases (downward jump) by 50 kN . Then, the V jumps from $+33.33 \text{ kN}$ to -16.67 kN . See the shear diagram for details.

Solutions to Extra Exercise #5, cont'd:

Step 2: Shear Diagram, Cont'd

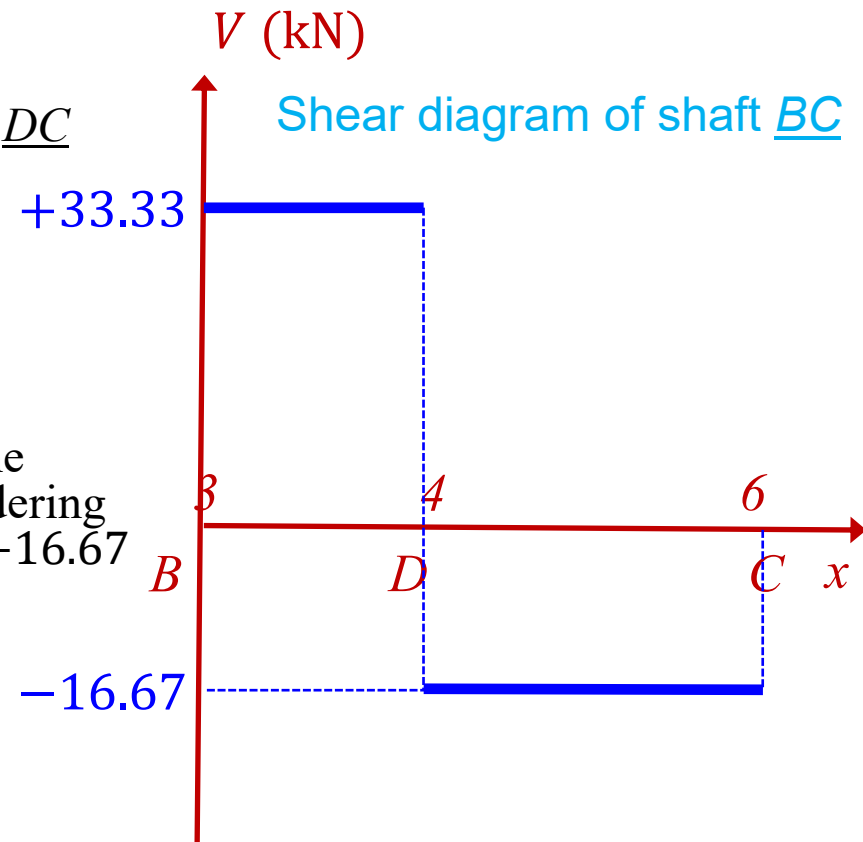
Note: we need to process the two beams separately!!!

For shaft BC:

d) Determine V diagram according to $\frac{dV}{dx} = w$ for $\underline{DC}$ region:

$$\begin{aligned}\frac{dV}{dx} &= 0 \\ \Rightarrow V &= C_3\end{aligned}$$

The constant C_3 can be determined by applying the “**continuity**” condition at last point D after considering the jump ($x = 1$, $V = -16.67$ kN). Thus, $C_3 = -16.67$ and $V = -16.67$ kN ($4 \text{ m} \leq x \leq 6 \text{ m}$).



Solutions to Extra Exercise #5, cont'd:

Step 3: Moment Diagram

For shaft BC:

a) Determine the moment at left end first (the origin B of x -axis):

$$x = 3, \quad M = 0$$

b) Determine $\mathbf{M}$ diagram according to $\frac{dM}{dx} = V$ for $\underline{BD}$ region:

$$\frac{dM}{dx} = 33.33$$

$$\rightarrow M = 33.33x + C_5$$

The constant C_5 can be determined by applying the “**boundary**” condition ($x = 3, M = 0$). Thus, $C_5 = -33.33$ and $M = 33.33x - 99.99 \text{ kN} \cdot \text{m}$ ($3 \text{ m} \leq x \leq 4 \text{ m}$).

Solutions to Extra Exercise #5, cont'd:

Step 3: Moment Diagram

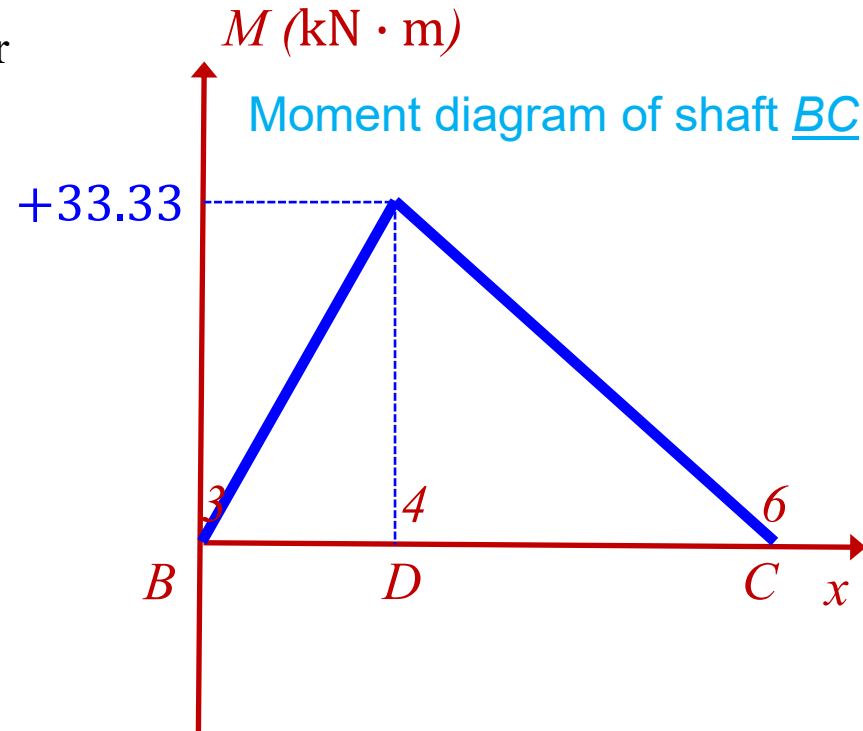
For shaft BC:

c) Determine **M** diagram according to $\frac{dM}{dx} = V$ for BD region:

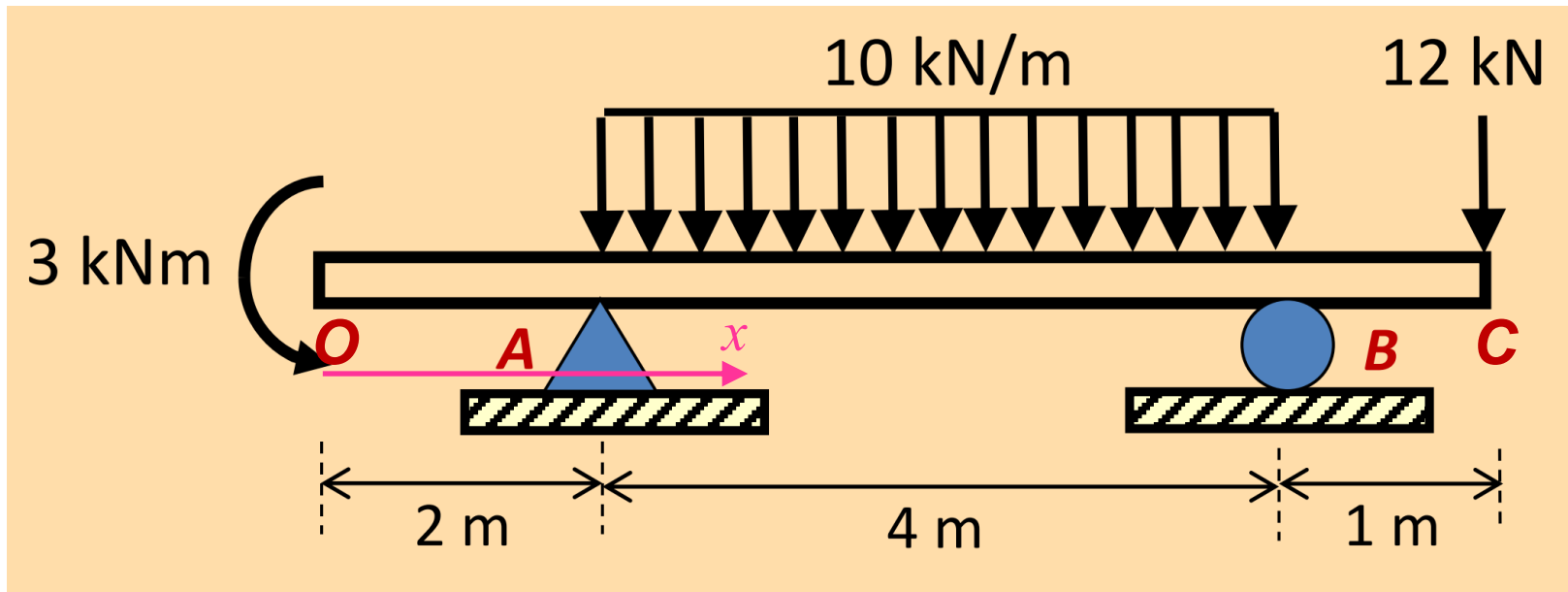
$$\frac{dM}{dx} = -16.67$$

$$\rightarrow M = -16.67x + C_5$$

The constant C_5 can be determined by applying the “**continuity**” condition ($x = 4$ m, $M = 33.33$ kN · m). Thus, $C_5 = 100$ and $M = -16.67x + 100$ ($4 \text{ m} \leq x \leq 6 \text{ m}$).



6. Determine the absolute maximum normal stress in the beam with external loading shown below. The beam has a uniform square cross-section with lateral size $a = 0.25$ m. Note: the shear and moment diagrams can be calculated either by section method or graphical method. (20')

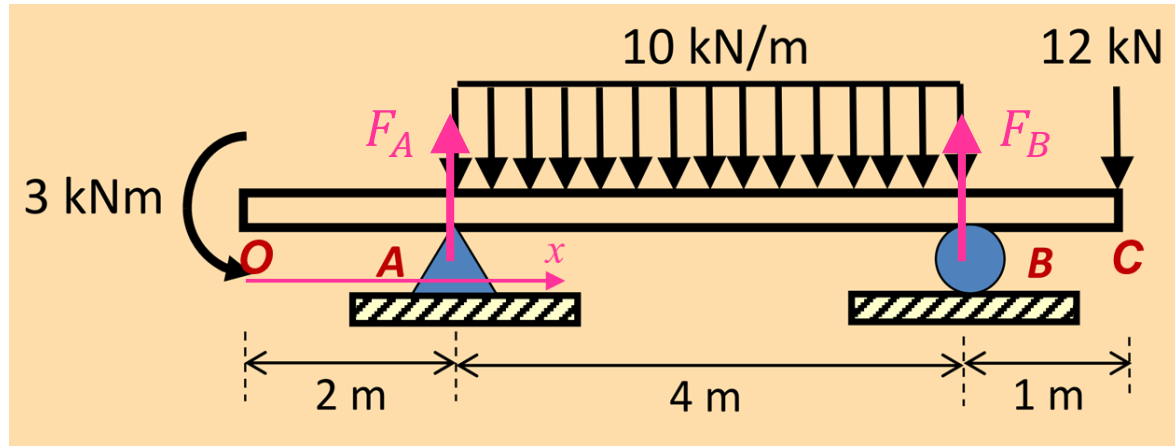


Solutions to Extra Exercise #6:

We use "Graphical Method" to solve this problem

Step 1: Support Reactions

Applying free-body diagram to the entire beam:



$$\sum F_y = 0 \rightarrow$$

$$F_A + F_B = 10000 \times 4 + 12000 = 52000$$

$$\sum M_A = 0 \rightarrow$$

$$F_B \times 4 + 3000 = 10000 \times 4 \times 2 + 12000 \times 5$$

$\rightarrow$

$$F_A = 17750 \text{ N}$$

$$F_B = 34250 \text{ N}$$

Solutions to Extra Exercise #6, cont'd:

Step 2: Shear Diagram

a) Determine the shear at left end first (the origin O of x -axis):

$$x = 0, \quad V = 0$$

b) Determine V diagram according to $\frac{dV}{dx} = w$ for OA region:

$$\frac{dV}{dx} = 0$$

$$\rightarrow V = C_1$$

The constant C_1 can be determined by applying the “**boundary**” condition ($x = 0$, $V = 0$). Thus, $C_1 = 0$ and **$V = 0$ ($0 \leq x \leq 2$)**.

c) At point A , due to the supporting reaction $F_A = 17750$ N (upward), the V increases (upward jump) by 17750 N. Then, the V jumps from 0 to +17750 N. See the shear diagram for details.

Solutions to Extra Exercise #6, cont'd:

Step 2: Shear Diagram, Cont'd

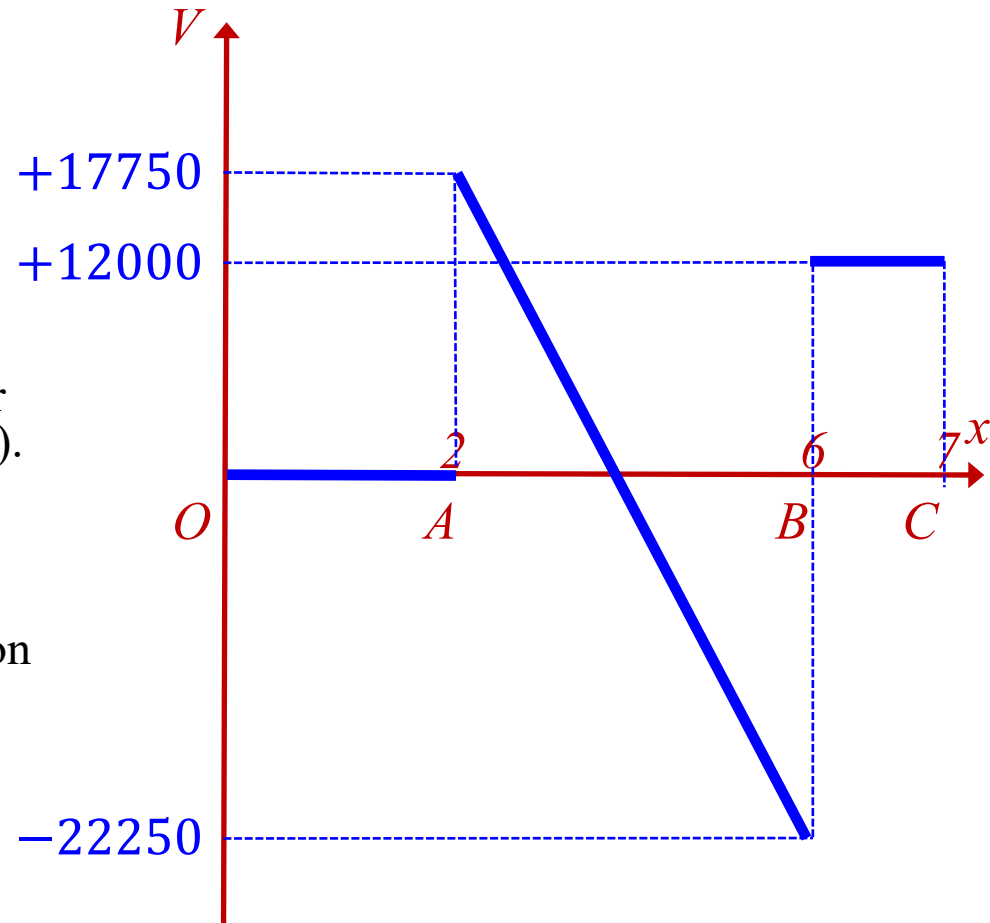
d) Determine V diagram according to $\frac{dV}{dx} = w$ for $\underline{AB}$ region:

$$\frac{dV}{dx} = w = -10000$$

$$\rightarrow V = -10000x + C_2$$

The constant C_2 can be determined by the “**continuity**” condition at last point A after considering the jump ($x = 2$, $V = 17750$). Thus, $C_2 = 37750$ and $V = -10000x + 37750$ ($2 \leq x \leq 6$).

e) At point B , due to the supporting reaction $F_B = 34250$ N (upward), the V increases (upward jump) by 34250 N. Then, the V jumps from -22250 to $+12000$ N. See the shear diagram for details.



Solutions to Extra Exercise #6, cont'd:

Step 2: Shear Diagram, Cont'd

f) Determine V diagram according to $\frac{dV}{dx} = w$ for $\underline{BC}$ region:

$$\frac{dV}{dx} = w = 0$$

$$\rightarrow V = C_3$$

The constant C_3 can be determined by the “**continuity**” condition at last point B after considering the jump ($x = 6$, $V = 12000$). Thus, $C_3 = 12000$ and **$V = 12000$ ($6 \leq x \leq 7$)**.

Solutions to Extra Exercise #6, cont'd:

Step 3: Moment Diagram

a) Determine the moment at left end first (the origin O of x -axis):

$$x = 0, \quad M = -3000$$

b) Determine **M** diagram according to $\frac{dM}{dx} = V$ for OA region:

$$\frac{dM}{dx} = V = 0$$
$$\rightarrow M = C_4$$

The constant C_4 can be determined by applying the “**boundary**” condition ($x = 0$, $M = -3000$). Thus, $C_4 = -3000$ and **$M = -3000$ ($0 \leq x \leq 2$)**.

Solutions to Extra Exercise #6, cont'd:

Step 3: Moment Diagram, Cont'd

c) Determine **M** diagram according to $\frac{dM}{dx} = V$ for AB region:

$$\frac{dM}{dx} = V = -10000x + 37750$$

$$\rightarrow M = -5000x^2 + 37750x + C_5$$

The constant C_5 can be determined by applying the “**continuity**” condition at last point A ($x = 2$, $M = -3000$). Thus, $C_5 = -58500$ and **$M = -5000x^2 + 37750x - 58500$** ($2 \leq x \leq 6$).

Solutions to Extra Exercise #6, cont'd:

Step 3: Moment Diagram, Cont'd

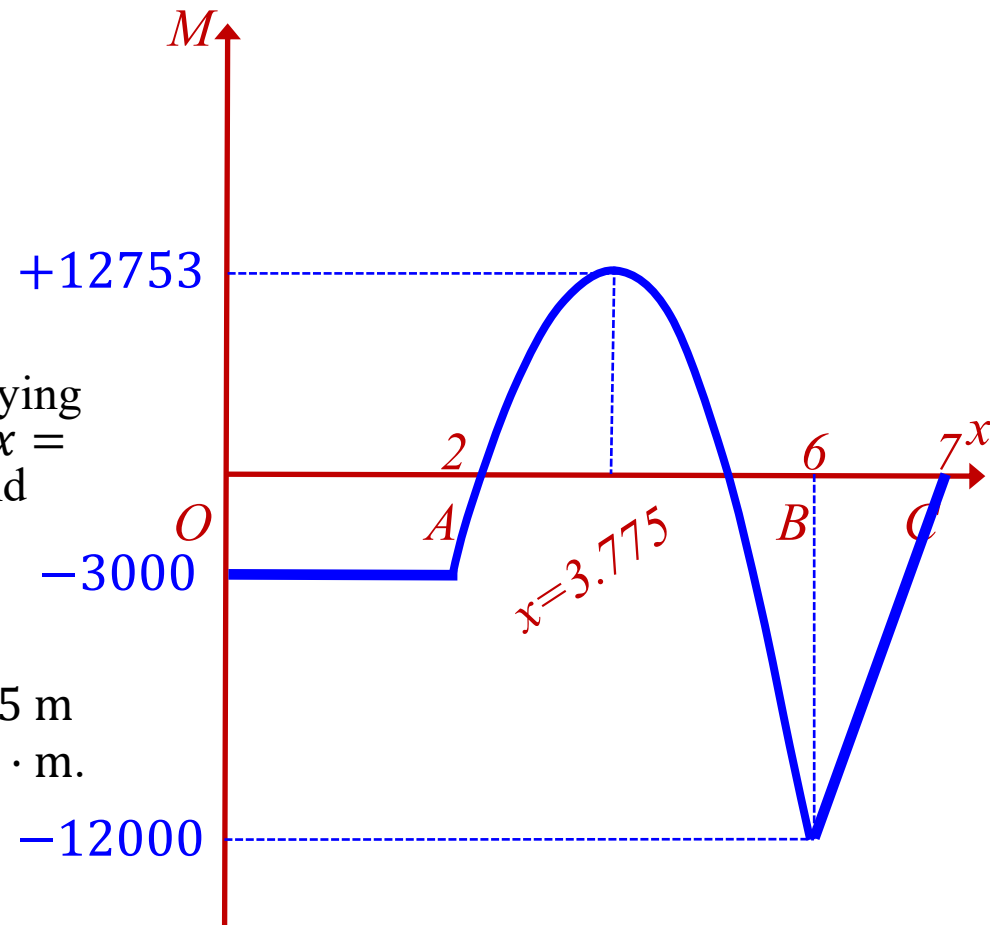
d) Determine **M** diagram according to $\frac{dM}{dx} = V$
for BC region:

$$\frac{dM}{dx} = V = 12000$$

$$\rightarrow M = 12000x + C_6$$

The constant C_6 can be determined by applying the “**continuity**” condition at last point B ($x = 6$, $M = -12000$). Thus, $C_6 = -84000$ and $M = -12000x - 84000$ ($6 \leq x \leq 7$).

The maximum moment occurs at $x = 3.775$ m
and the corresponding moment is $12753 \text{ N} \cdot \text{m}$.



Solutions to Extra Exercise #6, cont'd:

Step 3: Section Property

The moment of inertia I of sectional area about the neutral axis is (refer to the inside cover of the textbook for rectangular cross-section):

$$I = \frac{1}{12}bh^3 = \frac{1}{12} \times 0.25 \times 0.25^3 = 3.2552 \times 10^{-4} \text{ m}^4$$

Step 4: Maximum Normal Stress

The maximum normal stress is determined by:

$$\sigma_{\max} = \frac{Mc}{I}$$

$$\text{Thus, } \sigma_{\max} = \frac{12753 \times 0.25/2}{3.2552 \times 10^{-4}} = 4.90 \text{ MPa} \quad (\text{Ans.})$$